

## PHASE-CHANGE MEMORY

# Reducing the stochasticity of crystal nucleation to enable subnanosecond memory writing

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Operation speed is a key challenge in phase-change random-access memory (PCRAM) technology, especially for achieving subnanosecond high-speed cache memory. Commercialized PCRAM products are limited by the tens of nanoseconds writing speed, originating from the stochastic crystal nucleation during the crystallization of amorphous germanium antimony telluride ( $\text{Ge}_2\text{Sb}_2\text{Te}_5$ ). Here, we demonstrate an alloying strategy to speed up the crystallization kinetics. The scandium antimony telluride ( $\text{Sc}_{0.2}\text{Sb}_2\text{Te}_3$ ) compound that we designed allows a writing speed of only 700 picoseconds without preprogramming in a large conventional PCRAM device. This ultrafast crystallization stems from the reduced stochasticity of nucleation through geometrically matched and robust scandium telluride (ScTe) chemical bonds that stabilize crystal precursors in the amorphous state. Controlling nucleation through alloy design paves the way for the development of cache-type PCRAM technology to boost the working efficiency of computing systems.

**N**onvolatile phase-change random-access memory (PCRAM) is regarded as a leading candidate for next-generation electronic memory hierarchy (1–6). It uses the pronounced electrical resistance difference between the amorphous and crystalline states of chalcogenide phase-change materials (PCMs) to encode digital information (4). Large fragility—i.e., strong deviation of the temperature dependence of atomic dynamics from Arrhenius behavior (7)—is an essential property of PCMs that guarantees fast and reversible phase transitions between the two states at elevated temperatures, and yet good thermal stability at room temperature, making PCRAM one of the most promising candidates to compete with dynamic random-access memory (DRAM) and flash memory (8–10). However, to achieve “universal memory” (1) with PCRAM, subnanosecond operation is needed to compete with cache-type static random-access memory (SRAM) (5). Proper thermal design of PCRAM devices allows for an ultrafast RESET (erasing) process through amorphization,

while the SET (writing) process remains as the bottleneck, because the crystallization kinetics of PCMs is critically limited by their fundamental properties, such as nucleation rate and growth speed. Several strategies have been tried to improve the SET or writing speed of the current  $\text{Ge}_2\text{Sb}_2\text{Te}_5$  (GST)-based PCRAM devices by increasing the crystallization speed of GST, but the typical writing time is still tens of nanoseconds. A very fast SET speed of ~500 ps (11) was achieved on a ~30-nm pore-like GST-based PCRAM device with the aid of a constant low voltage. However, because a ~10-ns-long preprogramming treatment was needed before every SET operation, the real overall writing speed remains insufficient for subnanosecond cache-type memory applications.

The abundance of fourfold ABAB rings (A, Ge/Sb; B, Te) in the amorphous state (12, 13) has been proposed to be the mechanism for fast crystallization in GST, because the ABAB rings are the smallest structural units in the recrystallized cubic rock-salt phase, and two such structural motifs can form a cube. We call this structural order a crystalline precursor, because it shares the same feature of the corresponding crystalline structure. Crystalline precursors are related to subcritical embryos but do not necessarily imply the presence of quenched-in crystal nuclei. Upon heating to elevated temperatures (e.g., 600 K), atoms in the amorphous state become highly mobile, and rings and cubes fluctuate in and out constantly. A tremendous number of atomic configurations over a long incubation period need to be sampled before the critical nucleus is obtained (14–16) due to high frequency of formation and dissolution of crystalline precursors. Even in *ab initio* molecular dynamics simulations, the crystallization time at 600 K of models containing several hundreds of atoms can fluctuate

considerably, varying from several hundreds of picoseconds to many nanoseconds, reflecting the stochastic nature of the incubation process (16–19). To alleviate this, Loke *et al.* (11) introduced a long (~10 ns) pretreatment to preseed nuclei inside the amorphous matrix, such that the ensuing SET operation becomes primarily crystal growth. Instead, our goal is to achieve ultrafast crystallization by altering the intrinsic nucleation properties of the phase-change material itself to enable real subnanosecond memory writing.

Our design principle was to find materials with enhanced thermodynamic driving force to stabilize crystalline precursors—in this case, robust ABAB rings and cubes—to drastically extend their lifetimes during the incubation period. We also desire as much as possible geometric conformability between the crystalline precursor in the amorphous phase and the corresponding crystalline counterpart to reduce the interface energy. These two traits together—i.e., the dynamical stability and structural similarity—are projected to dramatically decrease the energy barrier for crystal nucleation (20, 21). To accomplish this goal, we introduced an alloying element to promote geometrically matched and high-strength chemical bonds to stabilize the crystal precursors. The most widely used PCMs are GeSbTe compounds along the pseudobinary line between GeTe and  $\text{Sb}_2\text{Te}_3$  (22, 23). Here, we used  $\text{Sb}_2\text{Te}_3$  as the parent phase to avoid the additional complexity of tetrahedral motifs found in amorphous GeTe and GeSbTe that arise from homopolar Ge-Ge bonds formed during fast quenching (24–26), because such motifs may hinder the crystallization into the octahedrally coordinated rock-salt phase. Antimony telluride is a prototype topological insulator with ordered quintuple layers connected via van der Waals forces, and it can also be made in a metastable rock-salt state (Fig. 1A) for phase-change application (27). Alloying elements such as transition metals into antimony telluride—e.g.,  $\text{Ti}_{0.4}\text{Sb}_2\text{Te}_3$  (TST)—can lead to superior crystallization speed as compared with GST (28), but the segregated triple-layered  $\text{TiTe}_2$  nanolamellae (29) prevent further reduction of crystallization time into the subnanosecond scale, stemming from the fact that none of the crystalline titanium tellurides match with the cubic rock-salt lattice structure of  $\text{Sb}_2\text{Te}_3$ .

Therefore, we performed a systematic materials screening of other transition metal tellurides (TMTs). We used two essential criteria to select the transition metal alloying element that best promotes high-fidelity crystalline precursors. The crystal-like structural motifs in the amorphous state should be (i) geometrically matched as much as possible to the rock-salt crystalline product  $\text{Sb}_2\text{Te}_3$  and (ii) further stabilized by the added transition metal alloying element if its incorporation brings in chemical bonds of high strength. The first criterion requires a local cubic geometry with a coordination number of 6 and bond lengths close to 3.0 Å (Fig. 1A). For the second criterion, we regarded the melting temperature ( $T_m$ ) and cohesive energy ( $E_{\text{coh}}$ ) as the key indicators. From all the TMTs with  $T_m > 900$  K

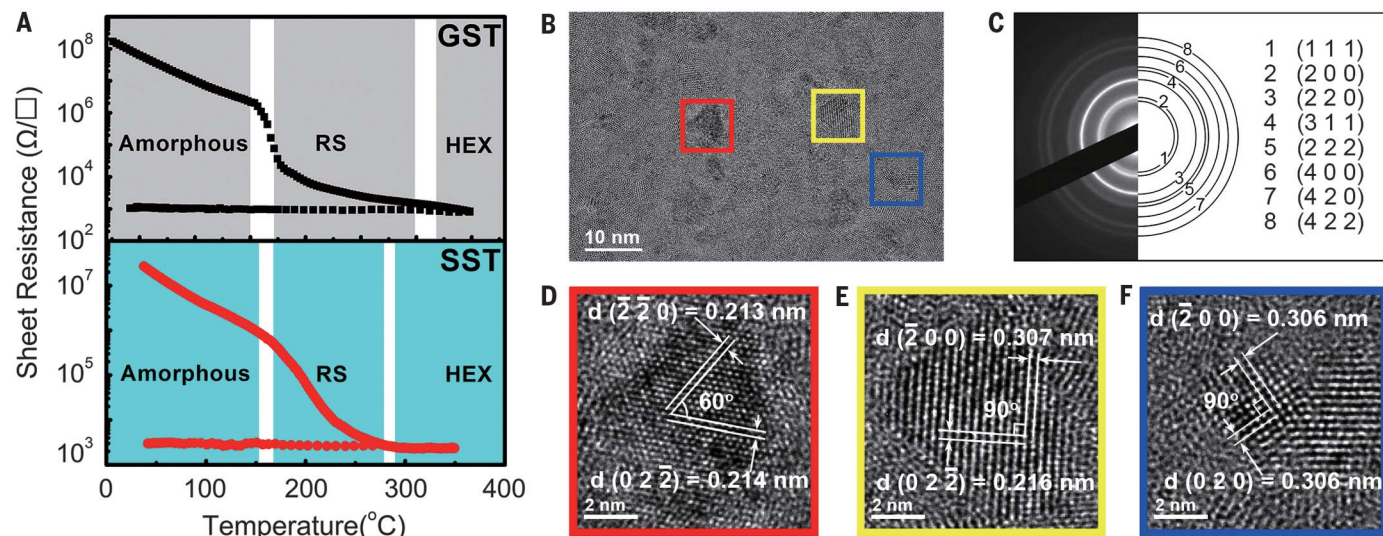
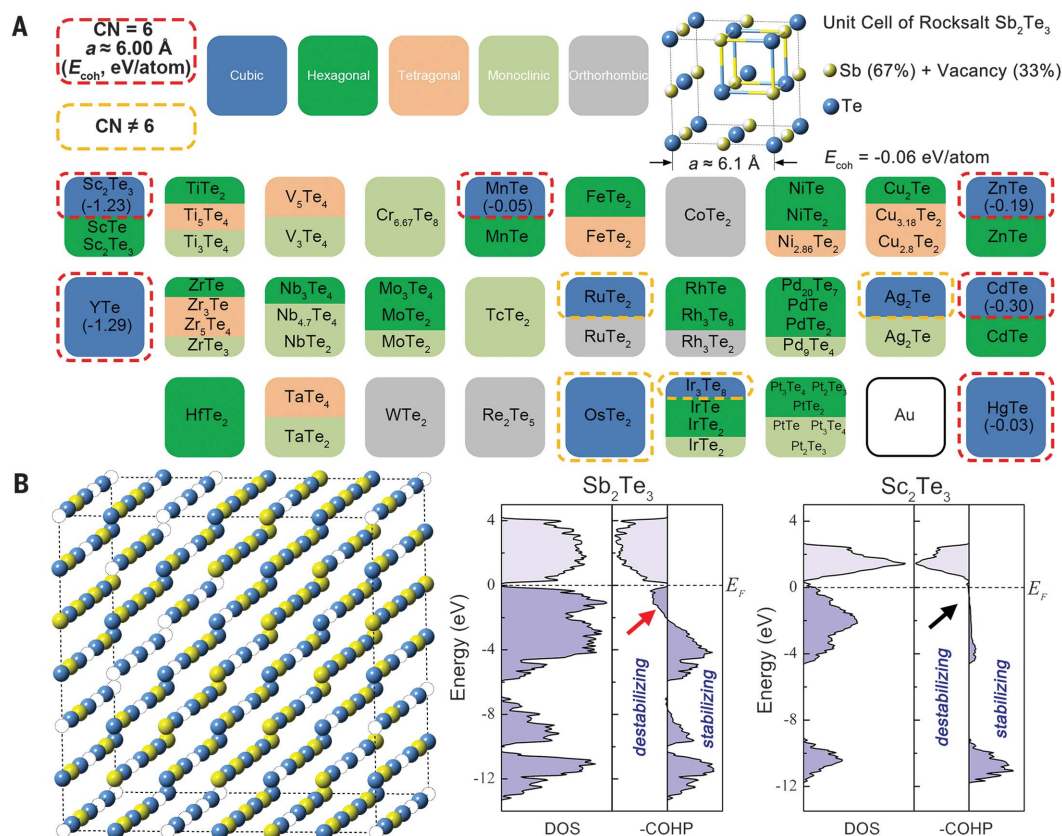
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**Fig. 1. Materials screening.**

(A) Screening geometrically matched TMTs with high-strength TM-Te bonds for rock-salt  $\text{Sb}_2\text{Te}_3$ . Only TMTs compounds with  $T_m > 900$  K are listed, in the format of the periodic table of elements. Six candidates (TM is Sc, Mn, Zn, Y, Cd, or Hg) with coordinate number CN = 6 and lattice parameter  $a \approx 6.00$  Å match closely with the rock-salt structure of  $\text{Sb}_2\text{Te}_3$ . (B) A 3 by 3 by 3 rock-salt  $\text{Sb}_2\text{Te}_3$  supercell model. Atomic vacancies, Sb and Te atoms are rendered with hollow circles and yellow and blue spheres. The left and right part of the -COHP curve indicates the antibonding (destabilizing) and bonding (stabilizing) interaction, respectively.



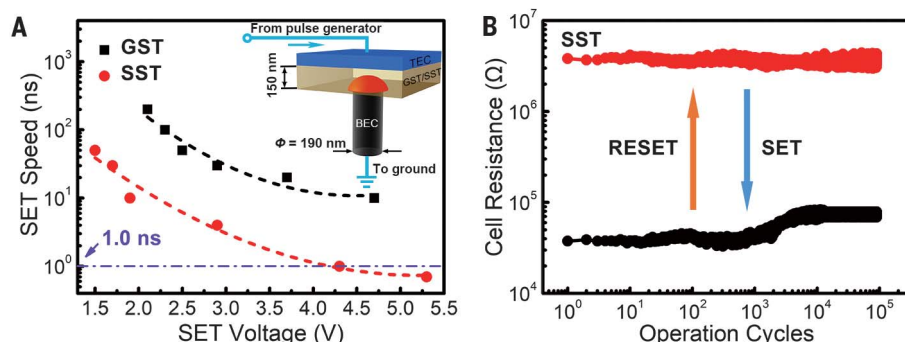
**Fig. 2. Rock-salt SST.** (A) Temperature dependence of the sheet resistance of ~300-nm-thick GST and SST films with the same heating rate of 10°C/min. Resembling GST, amorphous SST can be sequentially crystallized into metastable rock salt (RS) and equilibrium hexagonal

(HEX) phases. (B) TEM picture of ~20-nm-thick SST film annealed at 270°C. (C) The corresponding SAED pattern of (B). (D to F) High-resolution TEM images of three specific crystal grains framed in (B), project along  $[111]$ ,  $[011]$ , and  $[001]$  zone axes, respectively.

listed in Fig. 1A, we identify only six suitable candidates i.e., TM is Sc, Mn, Zn, Y, Cd, or Hg—that satisfied the geometric conformability criterion (fig. S1). Cubic rock-salt scandium telluride  $\text{Sc}_2\text{Te}_3$  is of particular interest, because it also has a high content of intrinsic atomic vacancies (1/3) on the

cation-like sublattice, the same as rock-salt  $\text{Sb}_2\text{Te}_3$  (Fig. 1B and fig. S1). We calculated the  $E_{\text{coh}}$  for the six rock-salt TMTs with density functional theory (DFT) simulations (30). The more negative value of  $E_{\text{coh}}$  corresponds to stronger TM-Te bonds (Fig. 1A). This ruled out MnTe and HgTe, as their

$E_{\text{coh}}$  is unfavorable with respect to that of base-alloy rock-salt  $\text{Sb}_2\text{Te}_3$  (−0.06 eV/atom). For the other four transition metals, DFT simulations on the rock-salt TM- $\text{Sb}_2\text{Te}_3$  revealed that Zn or Cd atoms resulted in too severe lattice distortions in the crystalline phase. Yttrium is not ideal because



**Fig. 3. PCRAM switching properties.** (A) Voltage dependence of the SET operation speed for SST and GST PCRAM devices with the same geometry. (Inset) Schematic of the device structure with the pulse signal applied to transform the phases in the mushroom-shaped active area right above the bottom electrode contact (BEC). (B) Cyclability: The SST device can repeatedly perform ultrafast SET (at 5.7 V) and RESET (at 7.5 V) operations up to  $10^5$  cycles with 800-ps pulses. The RESET and SET states are stable with sustainable resistance ratio.

the local motifs around Y atoms in the amorphous state can no longer keep the (defective-) octahedral coordination, which may hinder the crystallization kinetics (fig. S2).

Scandium is therefore singled out as the most appealing alloying element, according to the above criteria. We further cross-checked the chemical stability of  $\text{Sc}_2\text{Te}_3$  by using a more sophisticated method before synthesizing Sc-incorporated  $\text{Sb}_2\text{Te}_3$  alloys. We performed crystal orbital Hamilton populations (COHP) analysis, which dissects electronic density of states (DOS) into bonding (stabilizing) and antibonding (destabilizing) interactions (30). The antibonding contribution of both rock-salt  $\text{Sb}_2\text{Te}_3$  and  $\text{Sc}_2\text{Te}_3$  at the Fermi level  $E_F$  is marginal (Fig. 1B), indicating good stability of both systems. In  $\text{Sb}_2\text{Te}_3$ , we found an antibonding region right below  $E_F$ . In contrast,  $\text{Sc}_2\text{Te}_3$  had all the filled bands up to the  $E_F$  making stabilizing contributions, suggesting that  $\text{Sc}_2\text{Te}_3$  is more robust as compared with  $\text{Sb}_2\text{Te}_3$ . In this comparison both compounds are in exactly the same geometrical configuration (Fig. 1B), including the random distribution of atomic vacancies and lattice parameter, for the DFT simulations and COHP analyses (fig. S3).

Then we alloyed a small concentration of scandium into antimony telluride by magnetron sputtering (30). In general, too little scandium leads to poor thermal stability of the amorphous phase, whereas too much scandium makes our device fabrication difficult. Balancing the two led to the composition of  $\text{Sc}_{0.2}\text{Sb}_2\text{Te}_3$  (SST) for thorough experiments. The SST thin film that we obtained was amorphous, with a crystallization temperature of  $\sim 170^\circ\text{C}$  (Fig. 2A), similar to that of GST. We conducted electrical transport experiments on both SST and GST films upon heating, which gave similar sheet resistance profiles, as well as three resistance windows, the same as the GST, corresponding to three solid phases of SST, namely amorphous, cubic, and hexagonal (37). We performed transmission electron microscopy (TEM) experiments to access the detailed structural properties of the SST film. The bright-field (BF) image

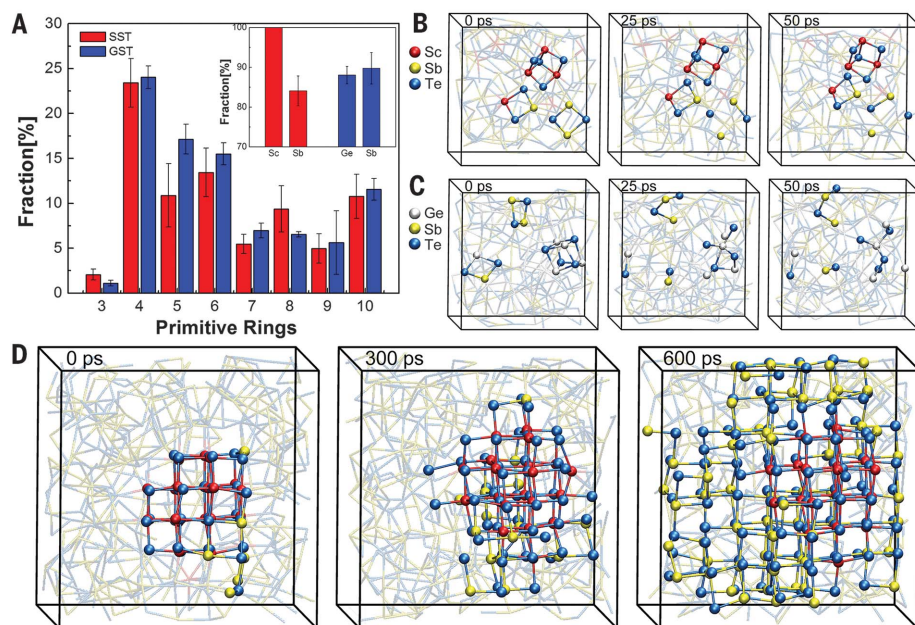
(Fig. 2B) and the corresponding selected-area electron diffraction (SAED) pattern (Fig. 2C) of the SST film annealed at  $270^\circ\text{C}$  show a homogeneously polycrystalline morphology with numerous nano-sized crystal grains, suggesting that SST is a nucleation-dominant material. The SAED pattern was indexed as rock-salt type, following the selection rules for crystal structures (32). The rock-salt structure was further confirmed with high-resolution TEM images (Fig. 2, D to F) of three (coexisting) specific crystal grains (boxed in Fig. 2B). In situ heating TEM experiments reveal the successive structural transformations from amorphous to rock salt and then to hexagonal phase in the SST film (fig. S4). Because the temperature window of rock-salt SST is comparable to that of GST, reversible and rapid phase transitions between the amorphous and rock-salt states are expected in SST-based PCRAM device through the design of a suitable heating profile. Just like GST, the equilibrium hexagonal phase needs to be avoided in the SET process because the transition to this phase is a relatively slow process, and the melting of the rigid hexagonal phase entails a high energy cost (27).

We fabricated SST-based T-shaped PCRAM devices by using  $0.13\text{-}\mu\text{m}$  node complementary metal-oxide semiconductor technology (inset of Fig. 3A) (30). GST,  $\text{Sb}_2\text{Te}_3$ , and TST devices of the same size were also made to enable a direct comparison. We altered the voltage pulses from nanosecond to picosecond width, with the magnitude ranging from  $\sim 1.0$  to  $\sim 5.5$  V, and applied them to the devices (fig. S5). As the magnitude of voltage pulse increases, the SET speed of all the devices becomes faster (Fig. 3A and fig. S6), with SST being one order of magnitude faster than GST at all voltages. The fastest SET process for the GST device needs  $\sim 10$  ns to complete, whereas that of the SST device requires only  $\sim 700$  ps. This SET speed is at the limit of PCRAM, because no preprogramming is needed for SST, and is already comparable to the resistance switching speed based on the so-called threshold switching effects (33). For the latter, the low-resistance state

disappeared instantly with electric field removal (33), whereas the transition in our SST is permanent, and the low-resistance crystalline state is very stable. The SST device showed a cyclability of  $\sim 10^5$  under the subnanosecond switching conditions (Fig. 3B). Up to  $\sim 4 \times 10^7$  cyclability was achieved by reducing the voltage bias while increasing the pulse width to tens of ns (fig. S7).

The subnanosecond crystallization speed of SST originates from the presence of the  $\sim 4\%$  Sc added into the  $\text{Sb}_2\text{Te}_3$  base alloy, because the fastest SET speed of the  $\text{Sb}_2\text{Te}_3$  device is  $\sim 6$  ns. For  $\text{Sb}_2\text{Te}_3$ , the absence of complexity through the introduction of Ge already improves the crystallization kinetics as compared with GST but is still insufficient to drive the SET speed down to the subnanosecond level (fig. S6). We performed DFT-based molecular dynamics (DFMD) simulations at finite temperatures (30) to elucidate the crystallization mechanism in SST and, in particular, the role of Sc. We studied the structural properties of the amorphous models, which we generated using the melt-quench scheme (22, 25). Fourfold rings are the dominant structural motif in amorphous SST (Fig. 4A), just like GST. More important, we found that every Sc atom was involved in at least one fourfold ABAB ring, whereas  $\sim 80$  to  $90\%$  of Sb atoms formed ABAB rings. This structural feature provides an essential ingredient for the high nucleation rate in SST. If the structural motif in the amorphous phase differs considerably from the crystalline phase, such as in the growth-driven PCM  $\text{Ag}_4\text{In}_3\text{Sb}_{97}\text{Te}_{26}$  (AIST), where fivefold rings dominate, the nucleation rate is very low (34, 35). This structural dissimilarity originates from both the parent compound  $\text{Sb}_2\text{Te}_3$  and the alloying Ag/In (35, 36).

We found that at an elevated temperature of  $\sim 600$  K, GST and SST show distinctly different behaviors: Ge(Sb)-Te-Ge(Sb)-Te rings break and reform frequently and rapidly with a very short lifetime on the order of  $\sim 5$  ps (fig. S8); in stark contrast, the robust Sc-Te-Sc-Te rings stay intact for over 50 ps (Fig. 4, B and C). The long lifetime of ScTe rings stems from the stronger ScTe bonds, and once two such rings get close to form a cube, the latter does not break easily (Fig. 4B and fig. S9). This behavior is obviously different from that of GST, where a sizeable crystalline cluster made of many connected cubes needs to form to prevent fast dissolution (14), and even an embedded crystalline seed containing as many as 58 atoms (in a 460-atom GST model) disappears rapidly at  $\sim 600$  K (16). Such an embedded seed had to be fully fixed, to enable quick crystallization through crystal growth in GST (37). In contrast, a crystalline precursor made of ScTe cubes ( $\sim 50$  atoms in a 4 by 4 by 4 SST supercell made of 428 atoms) can stand robust against thermal fluctuations at  $\sim 600$  K in the absence of artificial constraint, serving as the center for subsequent crystallization (Fig. 4D). In normal crystallization, a wide distribution of subcritical nuclei of varying sizes develops, and these embryos fluctuate in and out, with only a minute fraction of them evolving into critical nuclei (34, 38). In SST, the long lifetime of crystalline precursors



**Fig. 4. DFMD simulations.** (A) Primitive rings analysis of amorphous SST and GST. (Inset) Fraction of Sc and Sb atoms (respectively, Ge and Sb atoms) involved in at least one ABAB ring. (B and C) The stability of ABAB rings in SST and GST at ~600 K. (D) Crystallization process of SST with a crystalline embryo in the middle at ~600 K. This seed expands steadily and quickly with time to occupy much of the box within 600 ps, in contrast to the rapid dissolution of a similar-sized seed in GST (16). No constraint is applied to the ScTe seed during the DFMD simulations at ~600 K. The cutoff distance for bonds is chosen as 3.4 Å, corresponding to the first valley of the pair correlation function at the same temperature and is slightly larger than the maximum bond length (3.3 Å), determined by using sophisticated bonding analysis methods for GeTe/SbTe bonds at 0 K (26, 27, 41), to deal with thermal fluctuations at ~600 K.

due to the strong ScTe bonds allows for a quicker build-up of precursors on the verge of becoming nuclei or even quenched-in nuclei during phase-change operations, which leads to a superior SET speed in PCRAM devices. The particular ScTe seed that we introduced in our simulations is not necessarily already a nucleus above the critical size, because the latter is difficult to determine quantitatively from DFMD simulations because reasonable statistical sampling is computationally too expensive. Nevertheless, the size of the critical nucleus of SST should be smaller than that of GST, as the two compounds share the same interatomic distance and similar diffusion properties (the bulk diffusivity of both SST and GST is on the order of  $1 \times 10^{-10} \text{ m}^2/\text{s}$  at ~600 K), and yet SST crystallizes so much faster in our simulations (38).

To utilize SST for practical use, data retention at room temperature, RESET speed, and power consumption are also important aspects. The current composition  $\text{Sc}_{0.2}\text{Sb}_{2.8}\text{Te}_3$  contains ~4 atomic % of scandium, which is sufficient to guarantee a good thermal stability of the amorphous state as compared with GST (Fig. 2A). The data retention of the RESET state in SST device is estimated to be ~87°C for 10 years, very similar to that of GST (~82°C for 10 years) (fig. S10). This stability is due to suffocated diffusion in SST at low temperatures. However, scandium addition should not be excessive, so as not to make the growth kinetics too

sluggish at elevated temperatures for the desired crystallization. Regarding the RESET speed and energy, subnanosecond RESET operation was achieved in the SST device (fig. S6), and the RESET energy was one order of magnitude lower than in the GST device due to the easier melting of rock-salt SST (fig. S11). We believe that several strategies, such as scaling down the device size and fine-tuning the material composition, can improve the device performance further (39, 40) to rival SRAM, thus opening up the possibility to develop a truly universal memory. Our work is an example demonstrating the benefit of approaching the problem from materials design, taking advantage of known physical metallurgy principles to control nucleation and growth.

#### REFERENCES AND NOTES

- M. Wuttig, *Nat. Mater.* **4**, 265–266 (2005).
- F. Xiong, A. D. Liao, D. Estrada, E. Pop, *Science* **332**, 568–570 (2011).
- S. W. Nam *et al.*, *Science* **336**, 1561–1566 (2012).
- M. H. Lankhorst, B. W. Ketelaars, R. A. Wolters, *Nat. Mater.* **4**, 347–352 (2005).
- H. S. Wong, S. Salahuddin, *Nat. Nanotechnol.* **10**, 191–194 (2015).
- R. E. Simpson *et al.*, *Nat. Nanotechnol.* **6**, 501–505 (2011).
- C. A. Angell, *Science* **267**, 1924–1935 (1995).
- J. Orava, A. L. Greer, B. Gholipour, D. W. Hewak, C. E. Smith, *Nat. Mater.* **11**, 279–283 (2012).
- R. Jeyasingh *et al.*, *Nano Lett.* **14**, 3419–3426 (2014).
- A. Sebastian, M. Le Gallo, D. Krebs, *Nat. Commun.* **5**, 4314 (2014).
- D. Loke *et al.*, *Science* **336**, 1566–1569 (2012).

- J. Akola, R. Jones, *Phys. Rev. B* **76**, 235201 (2007).
- J. Hegedus, S. R. Elliott, *Nat. Mater.* **7**, 399–405 (2008).
- I. Ronneberger, W. Zhang, H. Eshet, R. Mazzarello, *Adv. Funct. Mater.* **25**, 6407–6413 (2015).
- G. C. Sosso *et al.*, *J. Phys. Chem. Lett.* **4**, 4241–4246 (2013).
- J. Kalikka, J. Akola, R. O. Jones, *Phys. Rev. B* **90**, 184109 (2014).
- J. Kalikka, J. Akola, R. O. Jones, *Phys. Rev. B* **94**, 134105 (2016).
- T. H. Lee, S. R. Elliott, *Phys. Rev. Lett.* **107**, 145702 (2011).
- J. M. Skelton, A. R. Pallipurath, T.-H. Lee, S. R. Elliott, *Adv. Funct. Mater.* **24**, 7291–7300 (2014).
- K. F. Kelton *et al.*, *Phys. Rev. Lett.* **90**, 195504 (2003).
- J. Russo, H. Tanaka, *Sci. Rep.* **2**, 505 (2012).
- N. Yamada, T. Matsunaga, *J. Appl. Phys.* **88**, 7020–7028 (2000).
- Z. Sun, J. Zhou, R. Ahuja, *Phys. Rev. Lett.* **96**, 055507 (2006).
- A. V. Kolobov *et al.*, *Nat. Mater.* **3**, 703–708 (2004).
- S. Caravati, M. Bernasconi, T. D. Kühne, M. Krack, M. Parrinello, *Appl. Phys. Lett.* **91**, 171906 (2007).
- V. L. Deringer *et al.*, *Angew. Chem. Int. Ed.* **53**, 10817–10820 (2014).
- Y. Zheng *et al.*, *Nano Res.* **9**, 3453–3462 (2016).
- M. Zhu *et al.*, *Nat. Commun.* **5**, 4086 (2014).
- F. Rao *et al.*, *Nat. Commun.* **6**, 10040 (2015).
- Materials and methods are available as supplementary materials.
- T. Siegrist *et al.*, *Nat. Mater.* **10**, 202–208 (2011).
- D. B. Williams, C. B. Carter, *Transmission Electron Microscopy A Textbook for Material Science* (Springer Press, New York, 2009).
- P. Zalden *et al.*, *Phys. Rev. Lett.* **117**, 067601 (2016).
- B. S. Lee *et al.*, *Science* **326**, 980–984 (2009).
- T. Matsunaga *et al.*, *Nat. Mater.* **10**, 129–134 (2011).
- W. Zhang, I. Ronneberger, Y. Li, R. Mazzarello, *Chem. Month.* **145**, 97–101 (2014).
- J. Kalikka, J. Akola, J. Larrucea, R. O. Jones, *Phys. Rev. B* **86**, 144113 (2012).
- K. F. Kelton, A. L. Greer, *Nucleation in Condensed Matter: Applications in Materials and Biology* (Elsevier, Oxford, 2010).
- G. Bruns *et al.*, *Appl. Phys. Lett.* **95**, 043108 (2009).
- I. S. Kim *et al.*, High performance PRAM cell scalable to sub-20nm technology with below 4F2 cell size, extendable to DRAM applications. *Symp. VLSI Tech. Dig.* **119b3**, 203–204 (2010). doi:10.1109/VLSIT.2010.5556228
- A. V. Kolobov, P. Fons, J. Tominaga, *Sci. Rep.* **5**, 13698 (2015).

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#### SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/358/6369/1423/suppl/DC1  
Materials and Methods  
Figs. S1 to S11  
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### Fast phase change with no preconditions

Random access memory (RAM) devices that rely on phase changes are primarily limited by the speed of crystallization. Rao *et al.* combined theory with a simple set of selection criteria to isolate a scandium-doped antimony telluride (SST) with a subnanosecond crystallization speed (see the Perspective by Akola and Jones). They synthesized SST and constructed a RAM device with a 700-picosecond writing speed. This is an order of magnitude faster than previous phase-change memory devices and competitive with consumer dynamic access, static random access, and flash memory.

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